# Saatvik Kesarwani

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## Education

**The University of Texas at Austin** **Expected May 2029**

*B.S. Electrical and Computer Engineering, Robotics Minor* | *Robotics Honors Program* | *GPA: 4.00/4.00* *Austin, TX*

- **Honors:** Grand Prize, AsterHacks (1/188); 3rd/828, HackVortex; Best Biomedical, HackTJ; National Merit Scholar
- **Relevant Coursework:** Data Structures, Multivariable Calculus, Linear Algebra, Computer Engineering, Physics I
- **Clubs:** IEEE UT, ML and Data Science Club, Texas Momentum, Texas Accelerate

**Thomas Jefferson High School for Science and Technology** | *AI, Machine Learning, Computer Vision* **Alexandria, VA**

## Experience

**Nesteye** · $10M-Valued AgTech Startup **Sept. 2026 – Present**

*Embedded Software Engineering Intern* *Austin, TX*

- Captured 500+ 3D poultry models at a $>$90% autonomous success rate by building an offline-first iOS LiDAR app
- Kept scans of moving birds accurate with ARKit motion gating that rejects frames with $>$100 mm motion residual

**Texas Aerial Robotics** **Oct. 2026 – Present**

*Hardware and Software Team Member* *Austin, TX*

- Enabled zero-risk pre-flight testing for a 4-drone IARC swarm by building a ROS-compatible Unity (C#) minefield simulator
- Planning safe swarm paths in under 10 min with a 1 ft mine buffer by developing AprilTag mine detection and mapping

**Dartmouth Hitchcock Medical Center, EDIT AI Program** **May 2025 – Aug. 2026**

*AI/ML Research Intern → Research Mentor* *Remote*

- Predicted survival for 397 colorectal cancer patients by fusing slide images, RNA-seq, methylation, and spatial data
- Ranked survival correctly for 72% of patient pairs (0.717 C-index) via gated attention fusion and TextGrad LLM loss
- Promoted to mentor 10 of 90 interns across 13 ML projects; presented at EDIT; 30-page paper selected for publication

**Obscurity Labs** **July 2025 – Aug. 2025**

*Software Engineering Intern* *Hybrid*

- Shipped 5+ UI features in 5 weeks by building the TypeScript front end of an endpoint detection and response platform
- Reduced bandwidth and backend storage by 80% with a Go log-deduplication agent; added 5+ real-time threat alerts
- Kept security event timelines accurate across regions by adding multi-time-zone support; demoed releases to the C-suite

**Orangewood Labs (YC-backed robotics startup)** **June 2024 – Aug. 2024**

*Forward-Deployed Engineering Intern* *Remote*

- Secured 10 product demos and pilot negotiations by mapping robot-arm use cases across 500+ factories and schools

## Projects

**AcousticTB** | *Python, TensorFlow, XGBoost, Arduino (C++), 3D printing* | GitHub **2024 – 2026**

- Screened tuberculosis from noisy cough audio at 0.85 AUC-ROC, surpassing WHO triage-test specificity targets
- Placed top 3 at 2 hackathons (1,000+ teams) with a self-designed handheld Arduino screener; sole author on IEEE TechRxiv

**OmniGlasses** | *Python, Dart/Flutter, OpenCV, YOLO, ESP32, FPGA* | GitHub **2026**

- Tracked eye strain at 98% blink/squint accuracy with an eye-state classifier that runs on the phone in our Flutter app
- Detected $\sim$95% of objects approaching from behind by turning FPGA collision events into instant spoken warnings
- Streamed live video from 3 ESP32 cameras to the phone over Wi-Fi by writing a custom UDP protocol that rebuilds frames

**Language-Controlled Robot Arm** | *Python, Flask, Three.js, Gemini API, Rotrics DexArm* **2025 – 2026**

- Hit 88% success in 50 robot-arm pick-and-place trials by having Gemini parse commands into grab, lift, and drop steps
- Blocked 100% of unsafe AI-generated paths in a Three.js simulation, then sent safe moves to the arm via a Flask REST API

**skadvising.org** | *React, TypeScript, Vite, TanStack Start* **2026**

- Drew 100+ client inquiries in 4 months by building, deploying, and hosting a React/TypeScript tutoring business site

## Skills

**Languages:** Python, C, C++, C#, Go, Java, TypeScript, JavaScript, Swift, Dart, SQL, CUDA, R, Assembly
**Libraries and Tools:** PyTorch, scikit-learn, OpenCV, YOLO, Hugging Face, pandas, FastAPI, Flask, React, Docker, Git, Linux
**Hardware:** Arduino, FPGA, I2C, oscilloscope, PCB design, sensor integration, bench testing, Fusion 360, Inventor
**AI-Assisted Development:** Claude Code, Codex, Cursor, agentic coding workflows, prompt engineering, AI code review